

Dongjun (Joyce) Xie

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SUMMARY

Applied AI engineer who turns ambiguous workflows into deployed products through customer-focused discovery, full-stack TypeScript/Python delivery, LLM tool integration, evaluation, and cloud operations.

EDUCATION

Northeastern University, San Jose, CA September 2022 - August 2025
Master of Science in Computer Science (**GPA: 4.0/4.0**)
University of California, Irvine, Irvine, CA September 2015 - March 2019
Bachelor of Arts in Quantitative Economics (**GPA: 3.5/4.0**)

TECHNICAL SKILLS

Programming Languages: TypeScript, Python, JavaScript, SQL, HTML/CSS.

Applied AI: LLM applications, agent orchestration, tool/API integration, RAG, evaluation, prompt and context engineering, streaming.

Frameworks: React, Next.js, React Native, Node.js, FastAPI, Flask, Hono, Vercel AI SDK.

Cloud & Data: AWS, GCP, Docker, Firebase, PostgreSQL, SQLite, Redis, Elasticsearch, DynamoDB.

PROFESSIONAL EXPERIENCE

SiriusMindshare LLC, San Jose, CA October 2025 - Present
Machine Learning Engineer | *Next.js, TypeScript, AWS, Vercel AI SDK*

- Translated ambiguous operating workflows into a production **5-agent AI product** for marketing, design, bookkeeping, support, and sales, with role-specific personas, tools, and memory.
- Owned full-stack delivery across **Next.js, TypeScript, AWS, and Vercel AI SDK**, including streaming UX, tool integration, observability, and cloud operations.
- Improved long-session reliability with **~4K-token prompt caching and 4-stage context compression**, reducing repeated inference work while preserving critical state.
- Applied the platform's agent-evaluation and multimodal reasoning approach to an **LVLMM retail attention-quality framework**; co-authored a paper accepted for publication and presentation at IEEE CCET 2026 (forthcoming).

1 Thing Against Racism Organization, Monroe, NJ

January 2025 - April 2025

Full-Stack Software Developer Intern (Remote) | *Python, Flask, Firebase, GCP*

- Built Flask/GCP APIs, caching, and real-time notifications for the Hate Crime Tracker; integrated social publishing workflows and improved **frequent-query response times by 40%**.

1 Thing Against Racism Organization, Monroe, NJ

January 2024 - April 2024

Full-Stack Software Developer Intern (Remote) | *TypeScript, React, React Native, Firebase*

- Delivered React and React Native experiences with Firestore synchronization, OAuth, encryption, and **AI-assisted content moderation** across iOS and Android.

PROJECTS

Agent Hook | Python, Webhooks, SQLite, Claude/Codex CLIs 2026

- Built a local-first trigger layer that converts cron, webhook, GitHub, and filesystem events into **bounded agent runs** with capability whitelists, retries, approvals, notifications, and audit history.
- Designed a provider-neutral execution model for Claude/Codex CLIs with **durable SQLite state and operator controls** for safe workflow automation.

In-Context Search | Chrome Extension, JavaScript, LLM APIs

2026

- Shipped select-to-explain and in-page search experiences combining **surrounding-page context with six AI/search providers** across inline, side-panel, and context-menu surfaces.

AgentFlow | TypeScript, React, Hono, WebSocket

2026

- Delivered a full-stack workflow product with **visual graph editing, multi-agent execution, and human gates** backed by REST/WebSocket APIs and pluggable LLM providers.

Additional open source: Agent Bridge (sandboxed agent compute) | jsfetch (JS-rendering CLI/MCP) | Kaggle Team (iterative agent team)